

AI CLUB · RESEARCH METHODS

Multi-Agent Research Assistant Demo

How can AI agents with different roles work together on research, and where do they go wrong?

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COORDINATION PATTERNS

Multi-agent systems use different coordination patterns

The architecture shapes who speaks, who decides, and where state lives.



CrewAI

Crews organize agent teams. Flows manage state, conditional logic, and event-driven workflows.



AutoGen

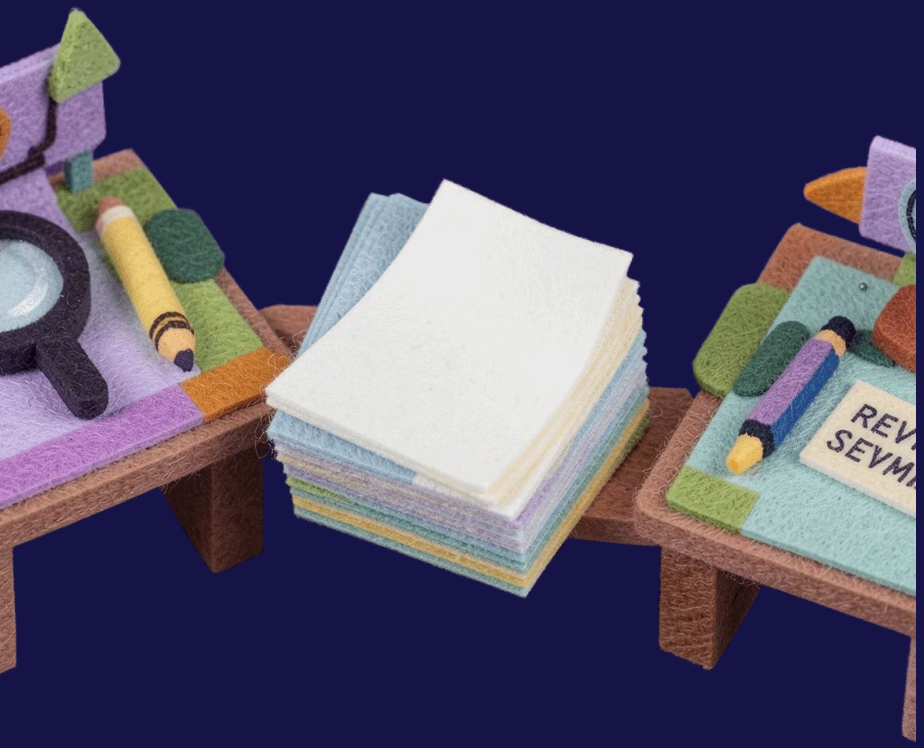
Multi-agent conversations use round-robin or model-selected speakers.



Direction of travel

AutoGen is in maintenance mode. Microsoft Agent Framework is the successor.

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THE DEMO SETUP

Three roles, one research task

Researcher

Collects facts and records sources

Writer

Produces a 300-word summary using only the notes

Reviewer

Fact-checks the summary and corrects it

We ran the roles manually with ChatGPT, Gemini, and Claude, then compared the result with a single-prompt baseline.

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FAILURE ANALYSIS

The pipeline preserved facts but lost trust signals

1 · Source names disappeared

The Writer dropped all source names from the summary.

2 · Tone became inflated

The Reviewer missed the omission, added hype words such as “critical” and “powerful,” and weakened attributions.

3 · Evidence narrowed

The Researcher cited one survey for several claims.

Core facts stayed correct

But the evidence trail became less visible, making the answer harder to trust.

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BASELINE VS. PIPELINE

More steps did not clearly beat one strong prompt

Role separation improved the research stage, but quality dropped at the Writer and Reviewer steps.

Dimension	Single prompt	Multi-agent
Speed	Faster	Slower
Source quality	Vague sources	Strong research step, then degraded
Unsupported claims	Some present	Core facts correct, evidence trail weakened
Overall result	Competitive baseline	No clear win

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TAKEAWAY

Reliability comes from controls, not agent count

Review every output

Human review is required for every output.

Constrain access

Use no private accounts; keep agents sandboxed.

Preserve the rules

Strict source, attribution, and tone rules must survive every handoff.

Conclusion: more agents do not automatically mean more reliability.

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